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Paleomagnetism of the Puente Piedra Formation, Central Peru

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Paleomagnetic samples were collected from 15 sites in the early Cretaceous Puente Piedra Formation near Lima, Peru. This formation consists of interbedded volcanic flows and marine sediments and represents the oldest known rocks of the Andean coastal province in this region. The Puente Piedra Formation is interpreted as a submarine volcanic arc assemblage which along with an overlying sequence of early Cretaceous clastic and carbonate rocks represents a terrane whose paleogeographic relationship with respect to the Peruvian miogeocline in pre-Albian time is unknown.

Moderate to high coercivities, blocking temperatures below 320°C, and diagnostic strong-field thermomagnetic behavior indicate that pyrrhotite is the dominant magnetic phase in the Puente Piedra Formation. This pyrrhotite carries a stable CRM acquired during an event of copper mineralization associated with the intrusion of the Santa Rosa super-unit of the Coastal Batholith at about 90 ± 5 m.y. B.P. The tectonically uncorrected formation mean direction of: $D = 343.2°$, $I = -28.6°$, $\alpha_{95} = 3.4°$ is statistically concordant in inclination but discordant in declination with respect to the expected direction calculated from the 90-m.y. reference pole for cratonic South America. The observed declination indicates approximately 20° of counterclockwise rotation of the Puente Piedra rocks since about 90 m.y. This is consistent with other paleomagnetic data from a larger crustal block which may indicate modest counterclockwise rotation during the Cenozoic associated with crustal shortening and thickening in the region of the Peru-Chile deflection.

1. Introduction

Paleomagnetic data from the western Andes of South America should provide constraints on the tectonic history of this very important orogen. Recent work in the North American Cordillera has revolutionized views of mountain-building processes and continental growth. It is now recognized that all basement terranes outboard of the Paleozoic-Mesozoic miogeocline have been added through magmatic or accretionary processes. More than 150 suspect terranes have been identified, most of which are oceanic in character and many of which are known to be allochthonous [1]. Paleomagnetism has proved very useful in interpreting the motion histories of these terranes [2,3].

Although there are numerous differences between the Andean and North American cordilleras, similar convergent plate interactions during the Mesozoic and early Cenozoic may have produced an accretionary collage along the western edge of South America. The purpose of this study was to identify an Andean suspect terrane and to investigate its motion history with respect to cratonic South America using paleomagnetism. Such a terrane was recognized near Lima, Peru whose early Cretaceous paleogeographic relationship with respect to cratonic South America is unknown. The oldest rocks of this terrane consisting of interbedded volcanic flows and marine sediments were sampled for paleomagnetic study. The characteristic remanent magnetization recorded in these rocks is considered to be a stable chemical remanence acquired during mid-Cretaceous intrusion of the Coastal Batholith of Peru. Comparison of the observed direction with an expected direction predicted from a mid-Cretaceous pole position for cratonic South America suggests a tectonic scenario

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of modest post mid-Cretaceous counterclockwise rotation and no significant latitudinal translation.

2. Regional geology

Four regional geotectonic elements are recognized in west-central Peru: a coastal belt of Mesozoic volcanic and sedimentary rocks, the late Cretaceous through mid-Tertiary Coastal Batholith, a belt of mid-Tertiary volcanic and volcaniclastic rocks associated with the late stages of the Coastal Batholith, and the highly deformed Mesozoic miogeoclinal sediments of the western Andes (Fig. 1).

Previous workers have attempted to correlate the early Cretaceous miogeoclinal sediments of the western Cordillera with rocks of similar age on the coast. However, lack of exposure under the mid-Tertiary volcanic belt and across the Coastal Batholith and extreme lithologic dissimilarity prohibit the synthesis of a convincing paleogeography. We consider many of the Mesozoic assemblages of coastal Peru which Cobbing et al. [6] refer to as "eugeosynclinal volcaniclastics" to be suspect. The paleogeographic relationship of these assemblages with respect to the Peruvian miogeocline during the early Cretaceous is uncertain. We recognize two tectonostratigraphic assemblages in the Lima area as suspect and refer to them collectively as the Congrejo terrane.

2.1. *Congrejo terrane*

The Congrejo terrane is a mildly deformed stratigraphic terrane composed of a Berriasian sub-

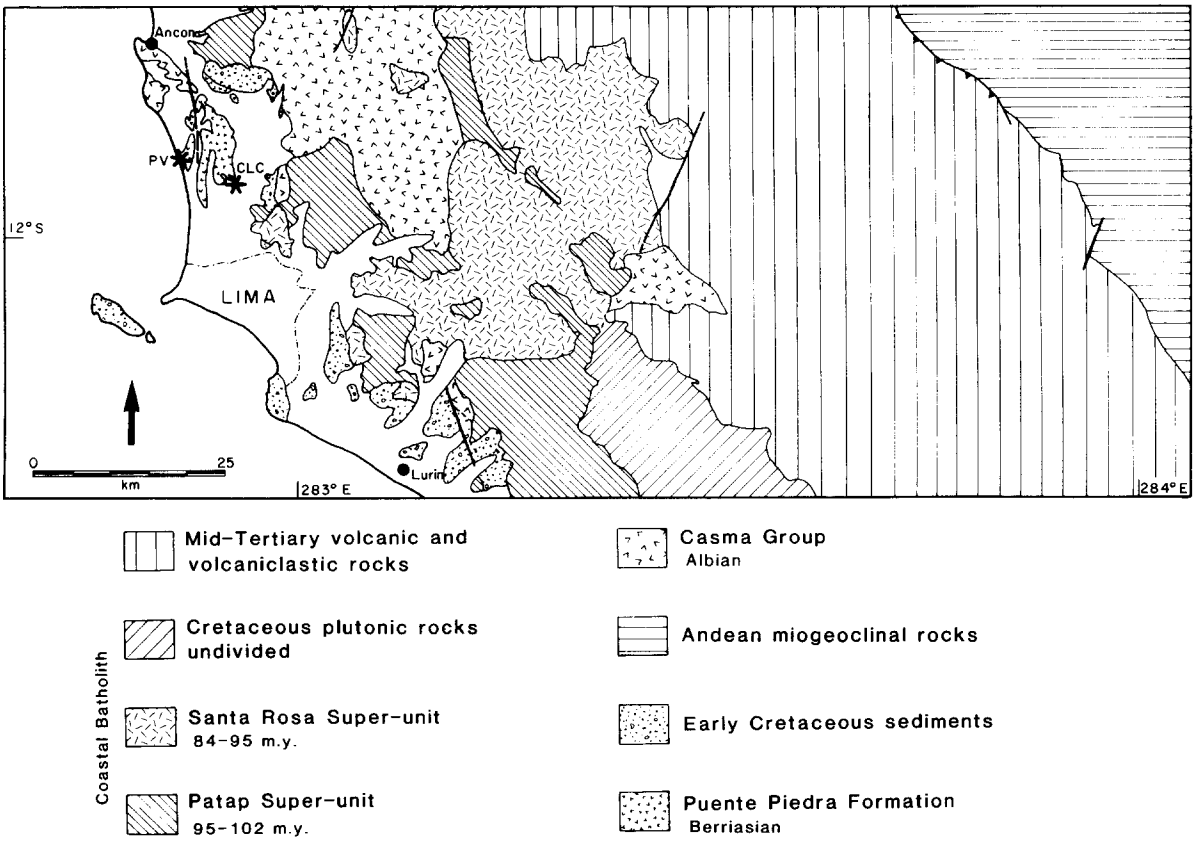


Fig. 1. Geologic map of the Lima area, west-central Peru modified from Megard [4]. Paleomagnetic localities: *CLC* = Cerro La Cuchilla, *PV* = Playa Ventanilla.

marine volcanic and sedimentary assemblage overlain by a Neocomian-Barremian(?) sequence of shallow marine and non-marine clastic and carbonate sediments. The basal assemblage includes the oldest known rocks in the central Peru coastal region, the Puente Piedra Formation (Figs. 1 and 2). This formation, composed of interbedded andesitic volcanic and marine sedimentary rocks, attains a thickness of 2000 m and was the main target for this study. It is not known what underlies the Puente Piedra Formation although gravity data suggest a crustal thickness of 30–40 km [7]. Surface outcrops are limited to a 250-km^2 area just north of Lima.

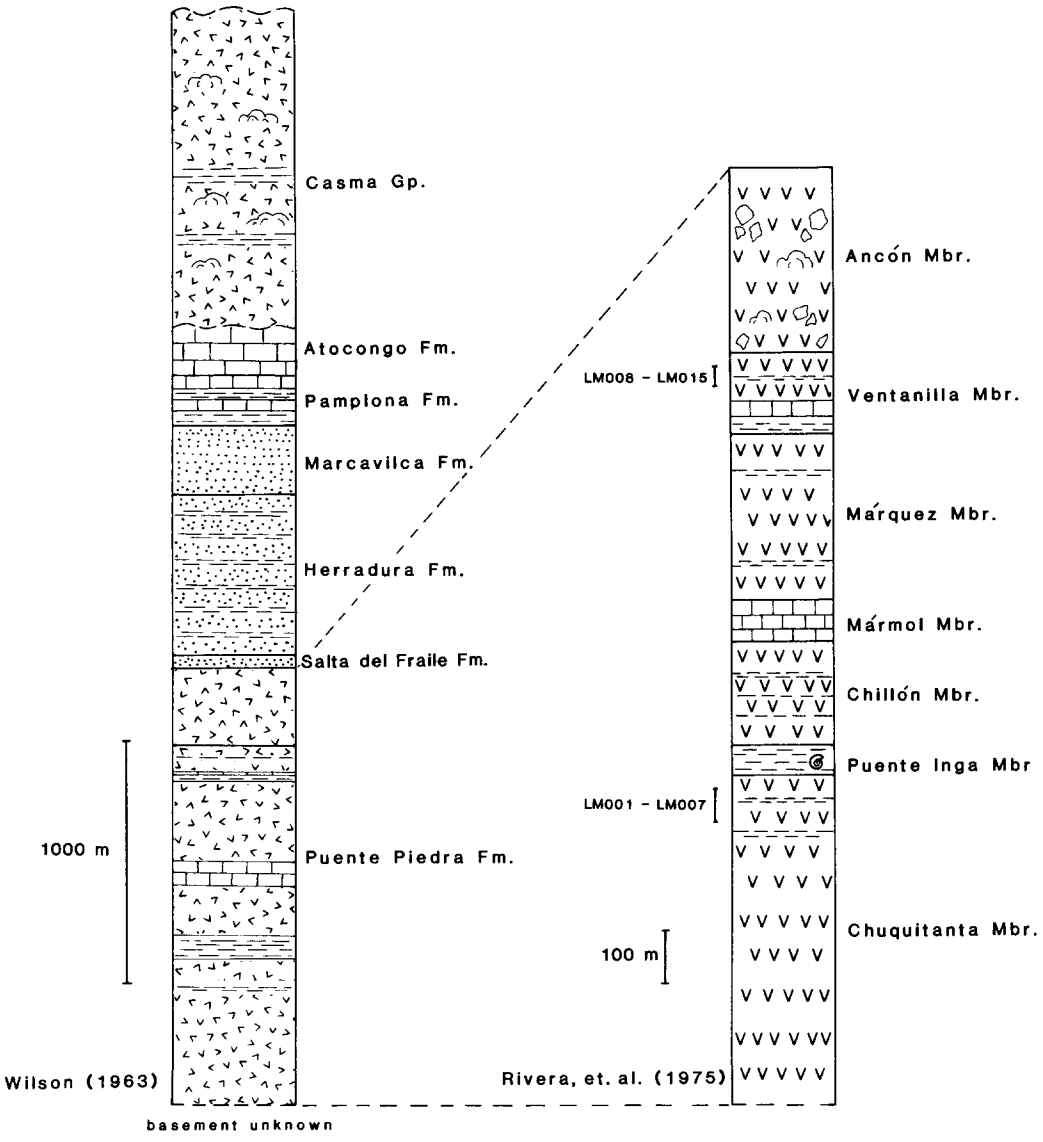


Fig. 2. Cretaceous stratigraphy of coastal central Peru from Wilson [9] and Rivera et al. [8]. The left column illustrates the stratigraphy of the Congrejo terrane and the Casma Group overlap assemblage. The right column shows the members of the Puente Piedra Formation and the stratigraphic position of paleomagnetic sites within the Chuquitanta and Ventanilla members.

The Puente Piedra Formation includes andesitic flows and tuffs, pillow lavas, marine shales and limestones and was subdivided into seven members by Rivera et al. [8] (Fig. 2). Shales of the Puente Inga member contain ammonites that Rivera et al. [8] considered to be Berriasian and probably late Berriasian. The Puente Piedra Formation represents a submarine volcanic arc whose paleogeographic relationship to cratonic South America in the early Cretaceous is unknown.

Overlying the Puente Piedra submarine volcanic arc assemblage is a series of early Cretaceous marine and non-marine sedimentary formations. These include the Salta del Fraile (= Morro Solar), Herradura, and Marcavilca Formations of probable Valanginian age and the Pamplona and Atocongo Formations of Hauterivian? and Barremian? age [9]. The lower three of these constitute a predominantly clastic sequence of non-marine and shallow marine subgreywackes and arenites attaining a thickness near Lima of 1014 m [9]. Paleocurrent indicators suggest ENE and SE flow directions and plutonic and volcanic clasts indicate derivation from some unknown source area to the west. These sediments differ significantly from the age correlative Chimu Formation of the Andean miogeocline which includes interbedded quartzose sandstone, shales, and thin coals with westerly to southwesterly paleocurrents [9]. The latter is interpreted as a westward prograding deltaic sequence.

The overlying Pamplona and Atocongo Formations consist of thinly bedded limestones and shales and thick bedded limestones respectively. The age of the Atocongo is poorly constrained although the Pamplona Formation contains late Valanginian or early Hauterivian molluscs [9].

2.2. *Overlap assemblage*

The lower Cretaceous sediments previously discussed are overlain unconformably by mid-Albian volcanic rocks and marine sediments of the Casma Group. Myers [10] showed that the Casma Group also rests depositionally on shales, quartzites, and quartzite conglomerates of the Andean miogeoclinal sequence. The Casma Group volcanics are therefore an overlap assemblage tying the Congrejo terrane to the Peruvian miogeocline by mid-Albian time. The uppermost units of the Casma volcanics are cut by 100-m.y.-old plutons of the Coastal Batholith [6]. West of the batholith, the Casma Group consists of thick andesitic tuffs with interbedded pillow lavas and marine sediments. Webb [11] suggests that early volcanics were deposited in fairly deep water but that the uppermost andesites were extruded onto subaerial stratovolcanoes. The Casma Group strata thin to the east where volcanic components become less dominant [6].

2.3. *The Coastal Batholith*

The Coastal Batholith is a linear composite chain of plutons from southern Ecuador to northern Chile and has been extensively studied by Cobbing and Pitcher [12], Cobbing [13], Pitcher [14], Cobbing et al. [15], and summarized in Cobbing et al. [6]. The Coastal Batholith ranges in age from approximately 100 m.y. to 35 m.y. and is composed of a number of compositionally and temporally distinctive "super-units".

Two important super-units are recognized in the Lima segment of the batholith (Fig. 1). The Patap, composed of gabbros and diorites, is the oldest super-unit of the batholith with age constraints of 95–102 m.y. [6]. The Santa Rosa super-unit is considered by Cobbing et al. [6] to be greater than 84 m.y. and less than 95 m.y. old and contains quartz diorite, tonalite, granodiorite, monzogranite and leucogranite in order of decreasing age. Subeconomic copper mineralization along the Lima segment of the Coastal Batholith is associated with the Santa Rosa super-unit [6]. These super-units intrude both the Congrejo terrane and Casma Group overlap and play an important role in the late Cretaceous remagnetization history of the Puente Piedra Formation.

2.4. *Structural history*

The main phase of deformation affecting Mesozoic sediments in the coastal region occurred in the late Albian, contemporaneous with the earliest magmatism in the Coastal Batholith and just slightly younger than the Casma Group over-

lap of the Congrejo terrane (Fig. 3). Strata of this terrane are folded into a gentle NNW-trending syncline and are cut by NNW-trending normal faults with minor stratigraphic displacement. The coastal Mesozoic assemblages are much less deformed than their temporal counterparts in the Andean miogeoclinal sequence. The timing of deformation in these two regions is also discordant as the main Andean orogeny was a late Cretaceous or Paleocene event [6]. Therefore, if the Congrejo terrane was allochthonous, its arrival by mid-Albian time resulted in only minor internal deformation and no deformation in the Andean region.

3. Paleomagnetic techniques

The two localities described in this study are located on the "Chancay" 1 : 100,000 scale topographic map (1982) available from the Instituto Geografico Nacional in Lima. Six to eight oriented cores were collected at each of fifteen sites at the two localities. Standard coring techniques were used and core orientations were made with a Brunton compass. Seven sites (LM001–LM007) were collected near Cerro La Cuchilla approximately 15 km north of Lima. Here a 75-m-thick sequence of interbedded volcanic flows and marine shales of the Chuquitanta member of the Puente Piedra Formation was sampled. These beds dip approximately 11° to the west and structural correction was made by rotating bedding to horizontal about the strike.

Eight sites (LM008–LM015) were collected at the south end of Playa Ventanilla approximately 7.5 km WNW of Cerro La Cuchilla. Here about 30 m of interbedded volcanic flows and marine sediments within the Ventanilla member of the Puente Piedra Formation were sampled. The structural attitude of these rocks was similar to that of the previous locality. Lithologies sampled include andesitic volcanic lavas, waterlain volcanic tuffs, and lithified mudstones.

Measurements of natural remanent magnetism (NRM) were done with a cryogenic magnetometer (ScT C-102) and alternating field demagnetization was done with a Schonstedt GSD-1 demagnetizer equipped with a tumbler. Thermal demagnetization was performed with a mu-metal shielded furnace which has a magnetic field less than 10 nT in the sample region. Thermomagnetic experiments were carried out using a Cahn 2000 microbalance interfaced with a microprocessor.

4. Rock magnetism

4.1. Thermomagnetic experiments

Magnetic separates were obtained by crushing the unweathered end cuttings of cores and extracting the magnetic fraction with a small permanent magnet. A strong field thermomagnetic balance was used to measure the apparent weight change of the sample in an applied field as the temperature was increased then decreased. All experiments were carried out in < 1 atmosphere of argon with a heating rate of 15°C/minute. Depending on the magnetization of each individual sample, the magnitude of the applied field ranged from 150 to 300 mT (1500–3000 Oe). J_{sat}-T curves observed in the thermomagnetic experiments are illustrated in Fig. 3.

The shape of the heating curve for samples LM014, LM015, and LM013 (Fig. 3A) is characteristic of two phase pyrrhotite as described by Schwarz [16], Schwarz and Harris [17], Schwarz and Vaughan [18], and Schwarz [19]. The increase in magnetization at 225°C represents a thermally induced reordering of cation vacancies in Fe_9S_{10} pyrrhotite and is known as the gamma transition [16]. The normally antiferromagnetic Fe_9S_{10} phase becomes ferrimagnetic at 225°C, thus the increased magnetization. The Curie temperature of the phase II pyrrhotite (Fe_9S_{10}) can be interpreted as that temperature where the trend of the initial heating curve intersects the heating curve beyond the gamma transition. In all three cases where a two-phase thermomagnetic result was observed, the phase II Curie temperature was approximately 290–295°C. This is 15–20° higher than results obtained by Schwarz [16]. The vacancy reordering that occurs at the gamma transition is not a reversible process. Therefore, the final room temperature magnetization of the two phase samples is greater than their initial value by a factor related

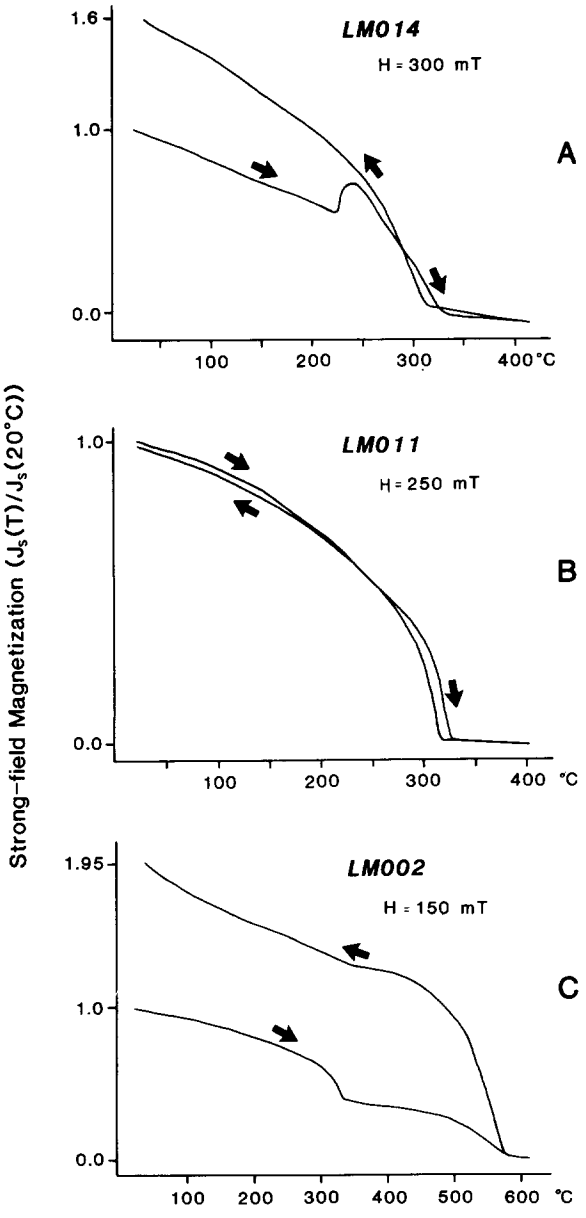


Fig. 3. Examples of strong-field thermomagnetic behavior for magnetic separates from the Puente Piedra Formation: (A) two-phase (Fe_7S_8 and Fe_9S_{10}) pyrrhotite, (B) single-phase (Fe_7S_8) pyrrhotite, (C) single-phase (Fe_7S_8) pyrrhotite with magnetite. Arrows indicate heating and cooling curves.

to the relative abundances of the two phases.

All samples exhibit a Curie temperature at approximately 320°C which is the Curie temperature of Fe_7S_8 pyrrhotite. In some sites such as LM011 (Fig. 3B) this phase is the only one present.

Along with pyrrhotite, sites LM002 and LM013 (Fig. 3C) were also found to have a 575°C Curie temperature phase indicative of magnetite. Neither of these sites showed significantly higher blocking temperature spectra for the NRM, but no stable direction was isolated in LM013.

A large increase in magnetization was observed in sites LM002 and LM013 which were heated to approximately 600°C. This is thought to reflect a breakdown of the pyrrhotite with loss of sulfur. Even though the experiments were performed in an argon atmosphere, the magnetic separates were not pure and contain some rock fragments. At high temperatures, the micro-environment of the sample capsule probably contains some free oxygen resulting in the oxidation of the iron. In other words, we believe that the heatings produced magnetite by high-temperature breakdown of the pyrrhotite followed by oxidation of the iron.

These thermomagnetic results are consistent with low blocking temperatures observed during thermal demagnetization and indicate that the dominant ferrimagnetic phase in the Puente Piedra rocks is Fe_7S_8 pyrrhotite. Ferrimagnetic pyrrhotite is also consistent with high coercivities observed during AF demagnetization and IRM acquisition experiments. Schwarz and Vaughan [18], report that saturation of randomly oriented pyrrhotite grains is not achieved until about 2000 mT (20,000 Oe) because of strong magnetocrystalline anisotropy.

The thermomagnetic results were corroborated by polished thin section microscopy. Pyrrhotite was observed both as disseminated grains and as microveinlet fillings. Euhedral pyrite grains along with minor amounts of chalcopyrite and covelite were also observed suggesting that the pyrrhotite was deposited during an event of Cu-Fe-S mineralization. Cobbing et al. [6] recognized an event of copper mineralization in coastal Peru which was associated with the intrusion of the Santa Rosa super-unit of the Coastal Batholith. Thus we believe that pyrrhotite precipitation was coeval with intrusion of the Santa Rosa super-unit and therefore consider the age of magnetization of the Puente Piedra Formation to be approximately 90

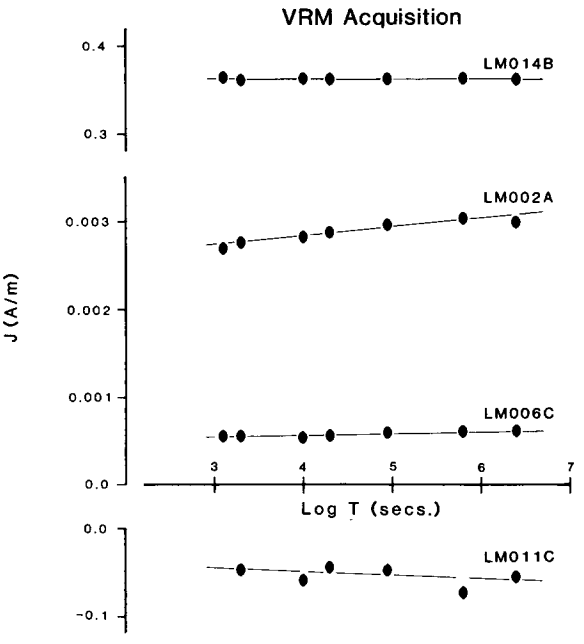


Fig. 4. Acquisition of viscous remanent magnetization. Data points represent magnetization acquired along the $+z$ axis of the individual cores. Total NRM intensities for these samples after isolation of characteristic directions were: LM014B—2.14×10^{1} A/m, LM002A—3.11×10^{-3} A/m, LM006C—5.98×10^{-4} A/m, and LM011C—3.08×10^{-1} A/m. (Note changes in scale.)

± 5 m.y. The observation of normal polarity for all sites is consistent with remagnetization during the Cretaceous normal polarity superchron (83–119 m.y.) [20].

4.2. *Viscous remanent magnetism*

Because the NRM directions of the Puente Piedra samples proved significantly different from the present field direction at the sampling locality and from an average Brunhes field direction, viscous remanent magnetism (VRM) was not considered to be a problem. Kligfield and Channell [21], however, describe significant VRM acquisition for limestones from the Helvetic Alps with pyrrhotite as the probable ferrimagnetic mineral. For this reason we conducted an experiment with the Puente Piedra rocks to evaluate their tendency to acquire VRM.

Four samples were selected after having been stored in a mu-metal shield for a number of months. The intensity of VRM acquired in an ambient field of 0.05 mT (0.5 Oe) was monitored as a function of time for periods ranging from 20 minutes to 8 weeks. The results shown in Fig. 4 indicate that no significant viscous remanence was acquired over this period.

5. Paleomagnetic results

5.1. *Natural remanent magnetism*

Intensities of natural remanent magnetism (NRM) ranged from 4.8×10^{-4} to 3.7×10^{-1} A/m. NRM directions were generally well grouped with north to northwest declinations and upward (negative) inclinations. Mean NRM directions do not differ significantly between the two localities although there is somewhat greater dispersion in declination at the Cerro La Cuchilla locality. The mean NRM direction of all stable sample samples in both geographic and stratigraphic coordinates is significantly different from the present geomagnetic field direction and the average Brunhes field direction at the sampling locality (Table 1).

5.2. *Alternating field demagnetization*

A representative sample from each of seven sites was subjected to progressive alternating-field (AF) demagnetization to peak fields of at least 80 mT (800 Oe). Vector demagnetization diagrams were used to evaluate the stability and coercivity spectra of the NRM.

Two samples showed little or no systematic response to AF demagnetization as illustrated in Fig. 5A. The direction of magnetization remains stable but the intensity is undiminished after cleaning in peak fields of 80–100 mT (800–1000 Oe).

Similar response is exhibited by three samples as shown in Fig. 5B. Although general decay of the magnetic vector towards the origin in the vector demagnetization diagram is observed, these samples still retain 50–60% of their initial NRM intensity after exposure to peak fields of 80–100 mT

TABLE 1

Paleomagnetic data

	$D(°)$	$I(°)$	$\alpha_{95}(°)$	N	R	k	Rock type
Site mean directions (thermally cleaned, tectonically uncorrected)							
LM001	−29.4	334.9	7.0	8	7.89	62.9	sediment
LM002	−25.6	343.5	10.3	6	5.88	43.1	sediment
LM003	−28.3	341.9	10.3	7	6.83	35.6	volcanic
LM004	−21.8	341.7	15.9	5	4.83	24.2	volcanic
LM005	−28.2	341.0	4.4	8	7.96	161.8	sediment
LM006	−26.5	348.8	15.2	4	3.92	37.6	volcanic
LM007	−31.5	355.1	9.9	5	4.93	61.0	sediment
LM008	−29.1	341.5	3.0	7	6.99	412.5	sediment
LM009	−31.2	338.5	5.5	3	3.00	509.6	sediment
LM011	−25.8	335.6	7.4	4	3.98	156.7	sediment
LM014	−29.6	350.7	10.3	5	4.93	56.3	sediment
LM015	−35.2	346.1	13.5	6	5.81	25.7	sediment
Formation mean directions (site latitude = −11.9, longitude = 282.9°E)							
uncorrected, uncleaned	−26.0	339.8	6.0	12	11.8	53.7	
uncorrected, cleaned	−28.6	343.2	3.4	12	11.93	167.4	
corrected, uncleaned	−25.5	345.3	6.1	12	11.79	51.9	
corrected, cleaned	−27.1	349.0	3.4	12	11.93	164.0	

Concordance/discordance statistics (from Beck [2] and Demarest [25]):

$R \pm \Delta R = -21.4 \pm 8.6°$ $F \pm \Delta F = 1.73 \pm 14.4°$

Present axial dipole field direction at site latitude and longitude: $I = -22.8°$, $D = 0.0°$.

(800–1000 Oe). The remanence of these three samples is composed of two components; one of low coercivity which was removed by demagnetizing fields of less than 10 mT (100 Oe) and one of high coercivity which is considered to represent the characteristic NRM direction.

Two samples from sites LM011 and LM013 showed significant response to AF demagnetization. LM011B maintains a relatively stable northwest-up direction through demagnetization in a peak field of 60 mT (600 Oe) at which time only 20% of the initial remanence remains (Fig. 5C). The magnetization of sample LM013E proved completely unstable during AF demagnetization and retained only 30% of its initial intensity after cleaning in a peak field of 30 mT (300 Oe).

5.3. *IRM acquisition*

Although a broad spectrum of AF demagnetization behavior was observed, it is clear that much of the characteristic remanence in the Puente Piedra Formation is of moderate to high coercivity. This conclusion was corroborated by isothermal remanent magnetism (IRM) acquisition experiments. Progressive field strengths between 50 and 700 mT (500–7000 Oe) were applied with the same orientation relative to the coordinate system of the sample. Because titanomagnetites do not exhibit coercive force greater than 300 mT, an increase in IRM above this magnetizing field indicates the presence of a higher coercivity phase [22].

Examples of IRM acquisition curves are shown in Fig. 6. They illustrate that significant amounts of IRM were acquired in the 300–600 mT range.

5.4. *Thermal demagnetization*

Single samples from each of eight sites were subjected to progressive thermal demagnetization in a program including peak temperatures of 115°, 170°, 250°, 280°, 320°, and 370°C. This demagne-

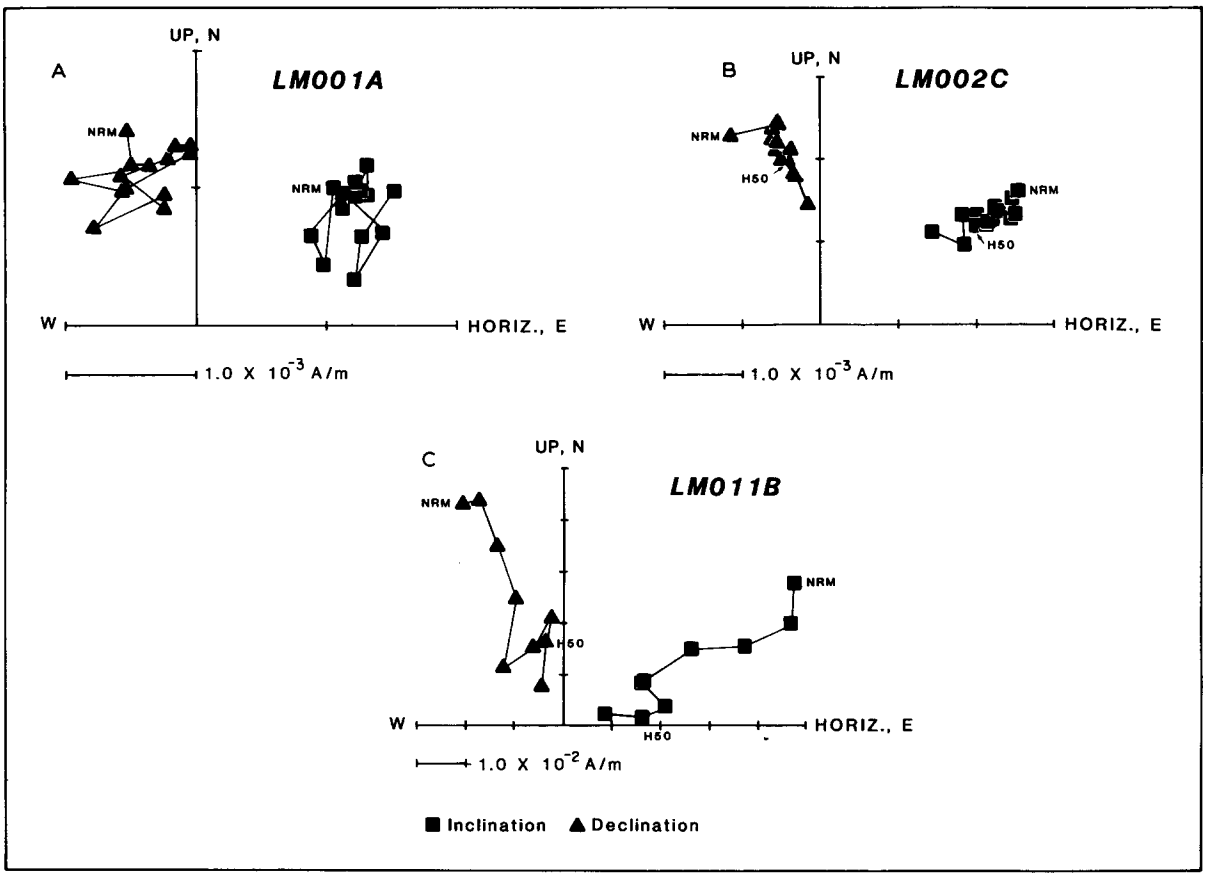


Fig. 5. Example vector demagnetization diagrams illustrating progressive AF demagnetization results. Bar indicates scale of each diagram. NRM and H50 next to data points indicate uncleaned direction and direction after 50 mT demagnetization step respectively.

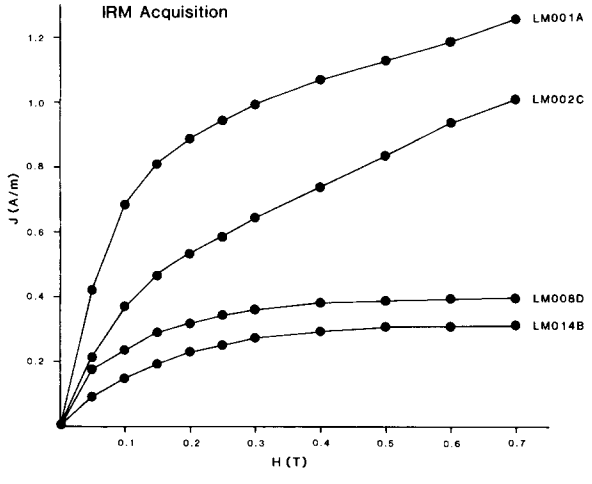


Fig. 6. Acquisition of isothermal remanent magnetization.

tization experiment confirmed the presence of blocking temperatures below 320°C and revealed that stable directions of magnetization could be isolated by demagnetization in the 200–320°C range. It was also established that variation in blocking temperature distribution between samples was sufficient to necessitate the use of multiple thermal steps for the blanket demagnetization of all remaining samples.

Blanket demagnetization of the Puente Piedra rocks included five progressive thermal steps of 185°, 220°, 270°, 290°, and 330°C. Directional changes and blocking temperature spectra were evaluated with vector demagnetization diagrams. The most common behavior during thermal demagnetization is illustrated in Fig. 7A and 7C.

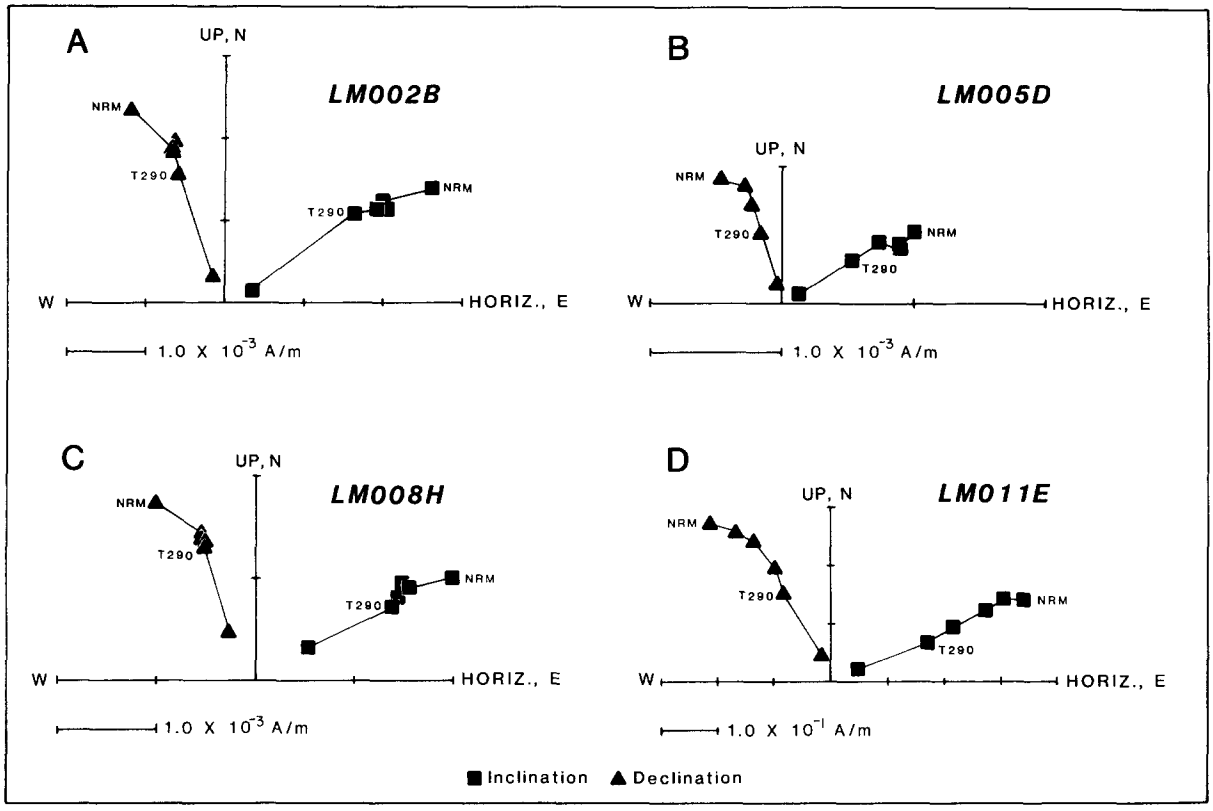


Fig. 7. Example vector demagnetization diagrams illustrating progressive thermal demagnetization results. Bar indicates scale of each diagram. NRM and T290 next to data points indicate uncleaned direction and direction after 290°C demagnetization step respectively.

The first thermal step (185°C) removes a low blocking temperature component. Higher steps isolate a stable direction with little decrease in intensity until the 270–290°C steps. The majority of blocking temperatures are distributed between the 290° and 330° steps.

A number of samples exhibited a much more even distribution of blocking temperatures between 0°C and 330°C as illustrated in Fig. 7B and 7D. These samples show the same secondary and characteristic components of magnetization as the previously discussed group.

The direction of magnetization remaining after the 290°C step was used from all samples to calculate site mean directions. Two samples from site LM004, three from site LM009 and all of sites LM010, LM012, and LM013 were deleted from the final data set because no stable direction was isolated during thermal cleaning. 68 samples from 12 sites were used in the final calculation of "cleaned" site mean directions.

6. Discussion and conclusions

6.1. Analysis of paleomagnetic results

The site mean and formation mean statistics were calculated using the techniques of Fisher [23] and are given in Table 1. In general, site mean α_{95} values decrease significantly after thermal cleaning and vary from 3.0° to 15.9° with a mean of 9.4°. Stereographic plots of tectonically uncorrected NRM and cleaned site and formation mean direc-

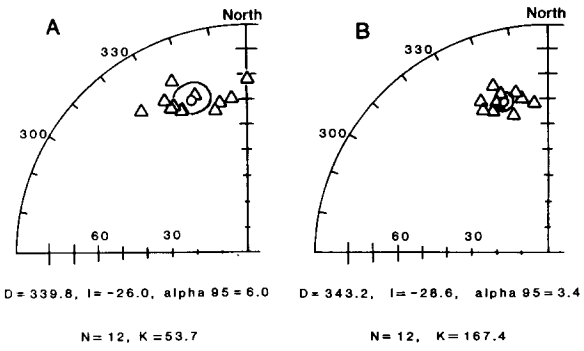


Fig. 8. Stereoplots of site mean (triangles) and formation mean directions with α_{95} for the formation mean: (A) uncleaned, tectonically uncorrected, (B) after 290°C demagnetization step, tectonically uncorrected. All directions are in upper hemisphere.

tions are shown in Fig. 8. Although the formation mean direction does not change significantly, the clustering of site mean directions is much tighter and the distribution of directions about the mean is more circular after thermal cleaning.

The precision parameter estimate (k) associated with the cleaned formation mean direction is high ($k = 167.4$) and could be taken to indicate failure to adequately average secular variation. However, we do not believe that this is the case. Our preferred interpretation is that the acquisition of magnetization through pyrrhotite mineralization is similar to the magnetization process of slowly cooled plutonic terrains. Precipitation of pyrrhotite from hydrothermal solutions generated by intrusion of the Santa Rosa super-unit may take considerable time, i.e. 10^3 to 10^5 years. The lifetime of the thermal envelope surrounding the Santa Rosa super-unit was at the very least 10^5 years and more probably 10^6 years. Physical conditions conducive for sulfide precipitation were driven by this heat source and undoubtedly evolved in both space and time. We entertain a scenario in which this physico-chemical potential field moved through the Mesozoic country rock in such a way as to

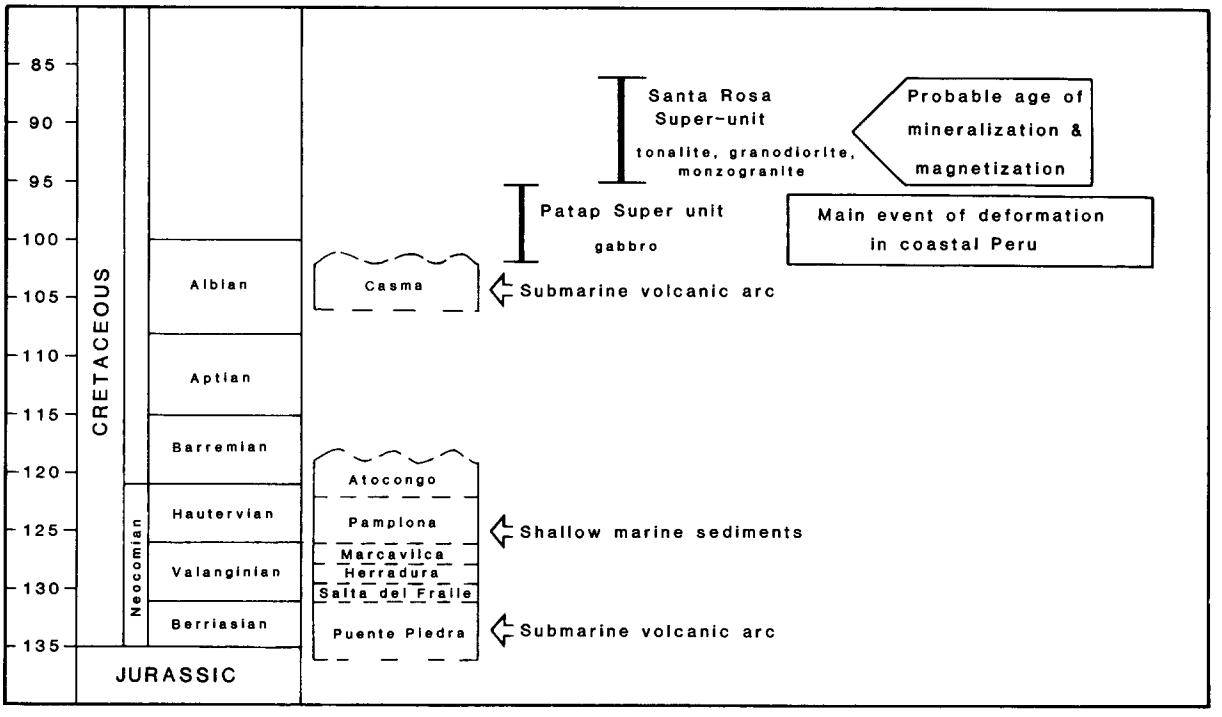


Fig. 9. Chrono-strato-plutono-magnetogram illustrating the ages and timing of important rock units and events for coastal central Peru during the Cretaceous.

produce mineralization and therefore magnetization over a significant duration with respect to the time frame of secular variation. Therefore, individual sites and perhaps even individual samples may have sampled the geomagnetic field for sufficient time to average secular variation. Considering the lack of confounding secondary components and the relatively low site mean k values we believe that the high formation mean k value is the result of within site averaging of much of the secular variation.

All rocks sampled dip gently to the WSW so that no fold test is possible. The formation mean k value does decrease after structural correction, however not significantly. Nevertheless, considering the regional structural history and the probable age of mineralization and magnetization, we believe that the Puente Piedra strata were folded prior to magnetization. Deformation of the coastal Peruvian Mesozoic rocks is considered by Cobbing et al. [6] to have been contemporaneous with intrusion of the Patap Super-unit of the Coastal Batholith (95–102 m.y.). These early phase gabbros are both deformed by and cut the family of structures which affect the mid-Albian and older units but which are always truncated by the 84–95 m.y. Santa Rosa plutons. The uncorrected formation mean direction is therefore the appropriate one to use for tectonic interpretations (Fig. 9).

6.2. *Tectonic implications*

An expected direction for the mean site location was calculated from the 90-m.y. reference pole (85°N, 41°E, $A_{95} = 10°$) given by Irving and Irving [24]. Comparison of this direction to the observed mean direction is made by way of concordance/discordance statistics summarized in Table 1. It should be noted that a typographical error in table 8 of Irving and Irving [24] lists mean poles as north when in fact the values given are south poles. Also, the 90-m.y. reference pole for South America is based on only four paleopoles, three from Argentina and one from Brazil. Rotation ("R") and flattening ("F") values were calculated by equations given in Beck [2] using the modification of Demarest [25].

The concordance/discordance analysis shows that moderate although significant counterclockwise rotation of the Puente Piedra rocks is indicated ($-21.4 \pm 8.6°$). A positive flattening value of 1.7° is statistically insignificant suggesting that there is no necessity for latitudinal translation of the Puente Piedra arc and overlying Cretaceous sediments since about 90 m.y. B.P.

Heki et al. [26] report similar results from "coastal volcanic rocks of Peru" of probable Cretaceous age. No information on the formations studied or on the magnetic properties is provided. A pole position based on 12 sites distributed over 10° of latitude and 5° of longitude at 68.0°S, 359.5°E, $A_{95} = 5.3°$ is concordant with the uncorrected Puente Piedra pole position at 73.3°S, 359.3°E, $dm = 3.7°$, $dp = 2.0°$.

Heki et al. [26] report modest counter-clockwise rotations for all Mesozoic units sampled north of the Peru-Chile deflection whereas rocks of Jurassic and Cretaceous age to the south in Chile yield concordant declinations. Whether this represents an oroclinal type rotation as Heki et al. [26] suggest or independent block rotations in Peru is unclear. The latter scenario would predict major left lateral strike-slip faults in coastal and Andean Peru which have not been described. Studies of modern seismicity generally indicate reverse faulting on NW-SE oriented structures concentrated along the eastern side of the Andean Cordillera [27].

Considering an oroclinal hypothesis, one can invoke a rotation pole at the north end, south end or in the center of the "Peruvian block". Both the southern and central Euler pole options would require hundreds of kilometers of extension in the Andean foreland at the northern end of the block. No evidence exists for this magnitude of Tertiary extension and seismicity clearly indicates a compressional regime. A northern Euler pole would predict significant compression of the crust along southern Peru, in northern Chile and in western Bolivia. Modeling based on gravity data indicates that crustal thickness in this area exceeds 70 km [7]. Crustal seismicity and late Cenozoic structures clearly indicate that active tectonics in the Peruvian Andes is characterized by deep-seated crustal shortening and thickening in equilibrium

with lateral growth in the foreland [27]. Andean tectonics is commonly associated with Nazca–South America plate interactions [28]; however, any model for the origin of the Andes must address the anomalous crustal thicknesses observed under the Peru-Chile deflection. Rotationally induced compressive stress accommodated by crustal shortening appeals to us as a possible scenario.

Using the crustal thickness data of James [7] for a cross section from the coast through Lake Titicaca and assuming an original crustal thickness of 35 km, we calculate approximately 325 km of crustal shortening. This amount of shortening corresponds to roughly 10–15° of counterclockwise rotation of the "Peruvian block" about a pole in northern Peru. This magnitude of rotation is consistent with our observations and with those of Heki et al. [26]. The available paleomagnetic data are insufficient to evaluate the timing of rotation compared to the history of crustal shortening determined from geological studies. Understanding the complex relationships between Nazca–South America plate interactions, intraplate rotation, and crustal thickening requires the collection of additional paleomagnetic data from rocks distributed throughout the Tertiary.

Paleomagnetic data from the Puente Piedra Formation do not allow us to constrain the motion history of the Congrejo terrane with respect to South America prior to 90 m.y. B.P. A scenario of pre-90 m.y. relative motion is allowable and seems likely based on geological considerations. This terrane must have docked by at least mid-Albian time as constrained by the Casma Group overlap and the plutonic stitching by the Coastal Batholith which followed shortly thereafter. This timing coincides with the main event of deformation in coastal Peru which was generally mild and more importantly only affected the region within 80–90 km of the present coastline. Arrival of the Congrejo terrane is undocumented in Andean miogeoclinal stratigraphy which is dominated by shales and limestones throughout the Albian. Intense deformation did not affect the Andes until late Cretaceous or Paleocene time and therefore cannot be related to the arrival of the Congrejo terrane. The Andes of central Peru are probably not the product of terrane collision and must be explained by some other convergent margin process.

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References

1 P.J. Coney, Accretionary tectonics in western North America, in: Relations of Tectonics to Ore Deposits in the Southern Cordillera, W. Dickinson and W. Payne, eds., Ariz. Geol. Soc. Digest, 14, 23–37, 1981.
2 M.E. Beck, Jr., Paleomagnetic record of plate margin tectonic processes along the western edge of North America, J. Geophys. Res. 85, 7115–7131, 1980.
3 S.R. May, P.J. Coney and M.E. Beck, Jr., Paleomagnetism and suspect terranes of the North American Cordillera, U.S. Geol. Surv. Open-File Rep. 83–799, 1983.
4 F. Megard, Etude géologique des Andes du Perou Central, Contribution A L'Etude géologique des Andes, 1, Mem. Orstom 86, 287 pp., 1978.
5 R. Rivera, La Fauna de los Estratos Puente Inga, Bol. Soc. Geol. Peru 22, 5–53, 1951.
6 E.J. Cobbing, W.S. Pitcher, J.J. Wilson, J.W. Baldock, W.P. Taylor, W. McCourt and N.J. Snelling, The geology of the western Cordillera of northern Peru, Inst. Geol. Sci., London, Overseas Mem. 5, 134 pp., 1981.
7 D.E. James, Andean crustal and upper mantle structure, J. Geophys. Res. 76, 3246–3271, 1971.
8 R. Rivera, G. Petersen and M. Rivera, Estratigrafia de la costa de Lima, Bol. Soc. Geol. Peru 45, 159–186, 1975.
9 J.J. Wilson, Cretaceous stratigraphy of central Andes of Peru, Bull. Am. Assoc. Pet. Geol. 47, 1–34, 1963.

10 J.S. Myers, Cretaceous stratigraphy and structure, western Andes of Peru between latitudes 10° and 10°30'S, Bull. Am. Assoc. Pet. Geol. 58, 474–487, 1974.
11 S. Webb, The volcanic envelope of the Coastal Batholith in Lima and Ancash, Peru, Ph. D. Thesis, University of Liverpool, 1976.
12 E.J. Cobbing and W.S. Pitcher, The Coastal Batholith of central Peru, J. Geol. Soc. London 128, 421–460, 1972.
13 E.J. Cobbing, The tectonic framework of Peru as a setting for batholitic emplacement, Pacif. Geol. 8, 51—62, 1974.
14 W.S. Pitcher, The Mesozoic and Cenozoic batholiths of Peru, Pacif. Geol. 8, 63–65, 1974.
15 E.J. Cobbing, W.S. Pitcher and W.P. Taylor, Segments and superunits in the Coastal Batholith of Peru, J. Geol. 85, 625–631, 1977.
16 E.J. Schwarz, Magnetic phases in natural pyrrhotite $Fe_{0.89}S$ and $Fe_{0.91}S$, J. Geomag. Geoelectr. 20, 67–74, 1968.
17 E.J. Schwarz and D.C. Harris, Phases in natural pyrrhotite and the effect of heating on their magnetic properties and composition, J. Geomag. Geoelectr. 22, 463–470, 1970.
18 E.J. Schwarz and D.J. Vaughan, Magnetic phase relations of pyrrhotite, J. Geomag. Geoelectr., 24, 441–458, 1972.
19 E.J. Schwarz, Magnetic properties of pyrrhotite and their use in applied geology and geophysics, Geol. Surv. Can., Pap. 74-59, 24 pp., 1974.
20 W.B. Harland, A.V. Cox, P.G. Llewellyn, C.A.G. Pickton, A.G. Smith and R. Walters, A Geologic Time Scale, Cambridge Earth Sci. Ser., pp. 63–82, Cambridge University Press, 1982.
21 R. Kligfield and J.E.T. Channell, Widespread remagnetization of Helvetic limestones, J. Geophys. Res. 86, 1888–1900, 1981.
22 D.J. Dunlop, Magnetic mineralogy of unheated and heated red sediments by coercivity spectrum analysis, R. Astron. Soc., Geophys. J. 27, 37–55, 1972.
23 R.A. Fisher, Dispersion on a sphere, Proc. R. Soc. London, Ser. A, 217, 295–305. 1953.
24 E. Irving and G.A. Irving, Apparent polar wander paths Carboniferous through Cenozoic and the assembly of Gondwana, Geophys. Surv. 5, 141–188, 1982.
25 H.H. Demarest, Error analysis for the determination of tectonic rotation from Paleomagnetic data, J. Geophys. Res. 88, 4321–4328, 1983.
26 K. Heki, Y. Hamano and M. Kono, Rotation of the Peruvian block from paleomagnetic studies of the central Andes, Nature 305, 514–516, 1983.
27 G. Suarez, P. Molnar and B.C Burchfiel, Seismicity, fault plane solutions, depth of faulting, and active tectonics of the Andes of Peru, Ecuador, and southern Columbia, J. Geophys. Res. 88, 403–428, 1983.
28 T.E. Jordan, B.L. Isacks, R.W. Allmendinger, J.A. Brewer, V.A. Ramos and C.I. Ando, Andean tectonics related to geometry of subducted Nazca plate, Geol. Soc. Am. Bull. 94, 341–361, 1983.